A3 High Energy Physics Coursework Theory Report

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Below, "we" is used instead of "I" to follow the convention of scientific writing.

0. Introduction

Symmetries are central to modern physics. Lorentz invariance underpins special relativity, while local gauge symmetries such as SU(2) govern the structure of the electroweak interaction. When neural networks are used to represent physical fields, a generic architecture does not automatically respect these symmetries and can only learn them approximately from data. Following the coursework, this report studies two pre-trained networks, NN1 and NN2, that approximate an SU(2) scalar doublet and a gauge–boson field respectively. We first verify their transformation properties analytically and numerically (Sections 1–2), then construct alternative architectures in which equivariance is enforced exactly by restricting the network input to Lorentz invariants (Section 3). Finally, we build a classical Lagrangian from these equivariant fields and confirm its invariance under both Lorentz and SU(2) transformations (Section 4).

1. Q1: Equivariance of NN1: SU(2) Fundamental and Lorentz Scalar

NN1 maps spacetime coordinates $x = (t, x, y, z) \in \mathbb{R}^4$ to four real outputs which we package as a complex SU(2) doublet $\Phi = (\phi_1 + i\phi_2,\ \phi_3 + i\phi_4)^T \in \mathbb{C}^2$.

Lorentz scalar (analytical). A scalar field satisfies $\Phi(\Lambda x) = \Phi(x)$ for all Lorentz transformations Λ. Unlike the SU(2) case this is a non-trivial constraint: it requires the network to depend only on Lorentz-invariant input combinations such as $x_\mu x^\mu = t^2 - x^2 - y^2 - z^2$.

Lorentz scalar (numerical). We test 500 random points under 50 random SO(3) rotations and Lorentz boosts, filtering for $|\Phi| > 0.1$ to avoid inflated relative errors. The mean relative errors are approximately 5.6% for rotations and 4.2% for boosts, and the max relative errors are both approximately 6.6%, confirming that NN1 is an approximate Lorentz scalar.

SU(2) fundamental (analytical). Under a global transformation $U \in \mathrm{SU}(2)$ the doublet transforms as $\Phi \to U\Phi$. Because SU(2) is an internal symmetry it acts only on the field components and not on the coordinates, so the doublet structure follows from how we package the outputs and places no constraint on the network weights. Unitarity ($U^\dagger U = I$) guarantees that $\Phi^\dagger\Phi$ is preserved.

SU(2) fundamental (numerical). We evaluate NN1 at sample points, apply random $U \in \mathrm{SU}(2)$ to the output doublet, and compare $\Phi^\dagger\Phi$ before and after. Differences are of order 10^{-9} (Table 1), consistent with float32 precision.

Table 1. SU(2) unitarity check for NN1: $|\Phi^\dagger\Phi - (U\Phi)^\dagger(U\Phi)|$ under random $U \in \mathrm{SU}(2)$.

Point	$\lvert\Phi\rvert^2$ before	$\lvert\Phi\rvert^2$ after U	$\lvert\Delta(\Phi^\dagger\Phi)\rvert$
0	1.028494	1.028494	9.01e-10
1	0.968938	0.968938	3.45e-08
2	0.071719	0.071719	1.66e-09
3	0.786423	0.786423	1.66e-09
4	0.815356	0.815356	1.73e-08

2. Q2: Equivariance of NN2: SU(2) Adjoint and Spin-1

NN2 maps spacetime coordinates $x \in \mathbb{R}^4$ to 12 real outputs, which we reshape into a 4×3 matrix $W_\mu^{\ a}(x)$ where $\mu = 0, 1, 2, 3$ is the Lorentz index and $a = 1, 2, 3$ is the SU(2) adjoint index.

Lorentz spin-1 (analytical). A Lorentz vector field transforms as $W_\mu^{\ a}(\Lambda x) = \Lambda_\mu^{\ \nu} W_\nu^{\ a}(x)$, i.e. the Lorentz transformation acts on the μ index while the SU(2) index a is left untouched. This is the defining property of a spin-1 field.

Lorentz spin-1 (numerical). We test 500 random points under 50 random rotations and boosts, filtering for $|W| > 0.1$. The mean relative errors are approximately 23% for rotations and 14% for boosts, and the max relative errors are approximately 28% and 27%, respectively. The error grows monotonically with the transformation angle (Figure 1), confirming that the network obeys the symmetry more accurately for small transformations.

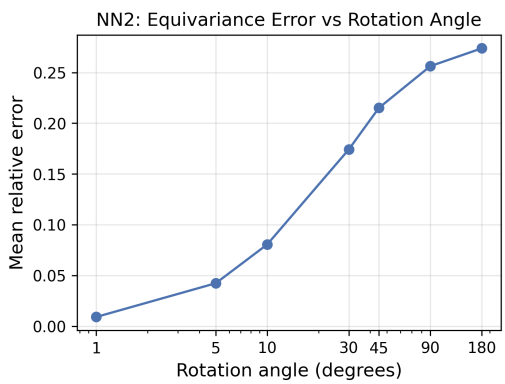

Figure 1. Relative error in $W_\mu^{\ a}(\Lambda x)$ vs transformation angle.

SU(2) adjoint (analytical). Under a global $U \in \mathrm{SU}(2)$ the adjoint representation acts via a real 3×3 rotation $R^{ab} = \frac{1}{2}\,\mathrm{Tr}(\sigma^a\, U\, \sigma^b\, U^\dagger)$, where σ^a are the Pauli matrices. The field transforms as $W_\mu^{\ a} \to R^{ab}\, W_\mu^{\ b}$. As with Q1, SU(2) is internal and does not act on the coordinates, so this transformation law follows from how we identify the second index with the Lie algebra basis. The matrix R is automatically a proper rotation ($\det R = 1$, $RR^T = I$), so the adjoint invariant $W_\mu^{\ a}\, W^{\mu\, a}$ is preserved.

SU(2) adjoint (numerical). We verify that $R \in \mathrm{SO}(3)$ and that $W_\mu^{\ a}\, W^{\mu\, a}$ is unchanged under random adjoint rotations. Differences are of order 10^{-8} (Table 2), confirming the invariant is preserved to machine precision.

Table 2. SU(2) adjoint invariance check for NN2: $W_\mu^a W^{\mu a}$ before and after adjoint rotation R.

Point	$W_\mu^{\ a} W^{\mu\, a}$ before	after R	$\lvert\Delta\rvert$
0	0.037461	0.037461	9.03e-10
1	2.396670	2.396670	4.13e-07
2	0.407864	0.407864	1.52e-08
3	2.395378	2.395378	4.02e-08
4	0.848923	0.848923	4.31e-08

3. Q3: Construction and Training of Equivariant Networks

The key idea is to build the symmetry into the network architecture so that it holds exactly, rather than relying on weight tuning to learn it approximately from data.

Analytical form.
We construct equivariant models of NN1 and NN2, denoted EquivariantPhi and EquivariantW, respectively.

EquivariantPhi. The only Lorentz-invariant scalar that can be formed from a single spacetime point is $s = x_\mu x^\mu = t^2 - x^2 - y^2 - z^2$. We therefore define

$$\Phi(x) = \mathrm{MLP}(s) \in \mathbb{R}^4. \tag{1}$$

Since s is invariant under Lorentz transformations, $\Phi(\Lambda x) = \mathrm{MLP}(s(\Lambda x)) = \mathrm{MLP}(s(x)) = \Phi(x)$ by construction. The four real outputs are packaged as an SU(2) doublet exactly as in Section 1.

EquivariantW. The only Lorentz 4-vector available from a single point is x_μ itself. The most general equivariant ansatz is therefore

$$W_\mu^{\ a}(x) = f^a(s)\, x_\mu, \tag{2}$$

where $f^a(s)$ is an MLP mapping $s \to \mathbb{R}^3$ (three adjoint components). Under a Lorentz transformation, $W_\mu^{\ a}(\Lambda x) = f^a(s)\,(\Lambda x)_\mu = \Lambda_\mu^{\ \nu}\, f^a(s)\, x_\nu = \Lambda_\mu^{\ \nu}\, W_\nu^{\ a}(x)$, which is the correct spin-1 transformation law, holding exactly by construction. The SU(2) adjoint structure is inherited from the index a as in Section 2.

Implementation and results.
Both MLPs have three hidden layers of 128 units with ReLU activations ($\sim$34k parameters each, compared to $\sim$50k for the originals). Training data is generated by evaluating NN1 and NN2 on 30,000 random points drawn uniformly from $[-5, 5]^4$, with 5,000 held out for validation. We train with Adam (initial learning rate 10^{-3}) and ReduceLROnPlateau scheduling for 150 epochs, selecting the checkpoint with the lowest validation MSE loss (Figure 2).

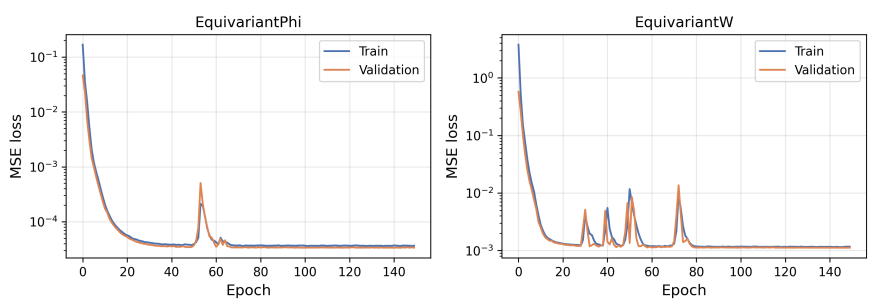

Figure 2. Training curves for EquivariantPhi and EquivariantW.

Verification of exact equivariance. We repeat the numerical tests from Sections 1–2 on the new architectures. Lorentz equivariance errors are now of order 10^{-7} (Table 3), consistent with float32 precision and confirming that the symmetry is exact by construction. This represents an improvement of five to six orders of magnitude over the original networks.

Emulation quality. On the validation set, EquivariantPhi achieves a mean relative error of 5.3% (median 3.5%), while EquivariantW gives 30.9% (median 15.7%). The higher error for W is expected: the ansatz in Equation (2) is the most general exactly equivariant form,

TABLE 3. Lorentz equivariance errors: original vs equivariant architectures.

Model	Transformation	Original	Equivariant
Φ	SO(3) rotation	0.0562	1.99e-07
Φ	Lorentz boost	0.0416	2.60e-07
W	SO(3) rotation	0.2302	3.45e-07
W	Lorentz boost	0.1415	4.80e-07

but the original NN2 is only approximately equivariant. The non-equivariant components of NN2 cannot be captured by an exactly equivariant network, so a residual emulation error is unavoidable.

4. Q4: Invariant Lagrangian Construction

Analytic form.
We construct a classical Lagrangian with kinetic and mass terms for both fields:

$$\mathcal{L} = \partial_\mu \Phi^\dagger \partial^\mu \Phi - m_\Phi^2\, \Phi^\dagger \Phi - \tfrac{1}{4}\, G^a_{\mu\nu}\, G^{\mu\nu\, a} + \tfrac{1}{2}\, m_W^2\, W^a_\mu\, W^{\mu\, a}, \tag{3}$$

where $G^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu$ is the Abelian field-strength tensor (we omit the non-Abelian $\epsilon^{abc} W^b_\mu W^c_\nu$ term for simplicity).

Lorentz invariance. Every Lorentz index is contracted with the Minkowski metric $\eta^{\mu\nu}$. Since Φ is a scalar and W^a_μ is a vector, each term is manifestly a Lorentz scalar.

SU(2) invariance. $\Phi^\dagger \Phi$ is invariant under $\Phi \to U\Phi$ by unitarity. The adjoint index a is contracted with δ^{ab}, which is invariant under $W^a \to R^{ab} W^b$ by orthogonality of the adjoint rotation R.

Implementation and results.
The equivariant networks from Section 3 output W^μ_a with an upper Lorentz index. We lower with the metric, $W^a_\nu = \eta_{\nu\mu}\, W^\mu_a$, before forming $G^a_{\mu\nu}$. All spacetime derivatives are computed exactly via PyTorch autograd. To ensure well-defined higher-order derivatives we retrain the equivariant networks from Section 3 with SiLU activations in place of ReLU, achieving comparable emulation quality (Figure 3). We set $m_\Phi = m_W = 1$.

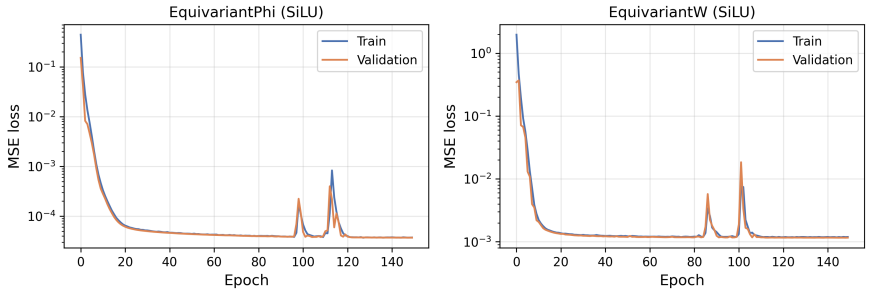

FIGURE 3. Training curves for SiLU equivariant networks.

Lorentz invariance (numerical). We evaluate $\mathcal{L}(x)$ and $\mathcal{L}(\Lambda x)$ at 15 random points under random SO(3) rotations and Lorentz boosts. The mean relative errors are 1.7×10^{-7}

and 2.7×10^{-7} respectively, confirming that the Lagrangian is a Lorentz scalar to machine precision.

SU(2) invariance (numerical). Since SU(2) is a global internal symmetry, we verify that the individual invariant combinations entering $\mathcal{L}$ are preserved under transformation. Under random $U \in \mathrm{SU}(2)$, the maximum deviations in $\Phi^\dagger\Phi$ and $W^a_\mu W^{\mu\, a}$ are 4.1×10^{-8} and 2.5×10^{-7} respectively, consistent with float32 precision. This is expected since the Lagrangian depends on the fields only through $\Phi^\dagger\Phi$ (unitary invariant) and δ_{ab} contractions (adjoint invariant), so SU(2) invariance holds as a mathematical identity.